CONNECTED COMPONENTS PULL BACK TO UNIONS OF CONNECTED COMPONENTS UNDER A FIELD EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author's opening sentence 'This is a purely topological statement.' is the first line of the proof and is reproduced as written.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-inverse-image-connected-component is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Continuous image of a connected set is connected. Standard topology, used at proof line 1116. Supplies that p(C) is a connected subset of X.
- Connected components are the maximal connected subsets. Standard topology, used at proof lines 1115-1117. Supplies that a connected subset of X meeting T lies in the connected component T.
- Base change of a scheme along a field extension. Standard, the formation of X_K with projection p: X_K to X. Supplies the object X_K whose components are being compared with those of X.

Lemma 0.1. *Let K/k be an extension of fields. Let X be a scheme over k. For every connected component T of X the inverse image $T_K \subset X_K$ is a union of connected components of X_K.*

Proof. This is a purely topological statement. Denote $p : X_K \to X$ the projection morphism. Let $T \subset X$ be a connected component of X. Let $t \in T_K = p^{-1}(T)$. Let $C \subset X_K$ be a connected component containing t. Then $p(C)$ is a connected subset of X which meets T, hence $p(C) \subset T$. Hence $C \subset T_K$. □

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.